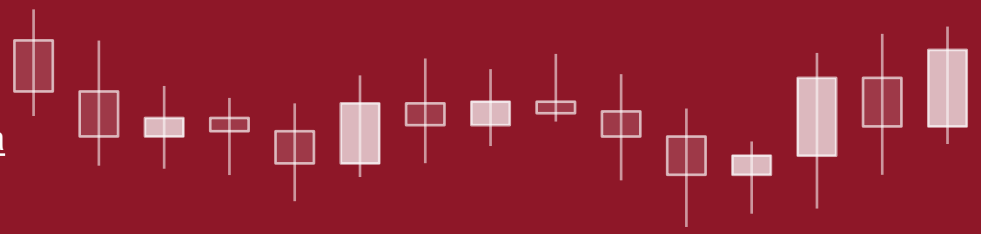
AI, MACHINE LEARNING AND BIG DATA FOR FINANCE AND MANAGEMENT · FINAL REPORT

Can Machine Learning Trade the Next Nasdaq Candle?

Next-candle direction models and long/short backtests on E-mini Nasdaq-100 futures (NQ), after commissions

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Waseda University · November 2026
Option 1: data analysis · Code and results: github.com/Paciooo/Project_ai_ML_bigdata



Executive Summary

Models predict slightly better than chance; none of the five validation-selected candidates passes every robustness test

6.2M

one-minute NQ bars (2006–2025), tested on 1, 5, 15, 60-minute and daily candles

+0.62 pp

best gain over “always up” among validation-selected models (MLP 128-64-32, 5 min; exploratory 95% CI 0.32 to 0.92)

407

model × direction × filter strategies compared on validation data

0 of 5

validation-selected candidates passing every pre-set test (out of sample, after costs, walk-forward)

Problem

Can a model predict whether the next Nasdaq futures candle closes above its open, and trade that signal profitably after $5 round-trip commissions?

Approach

Original vs optimised logistic regression, XGBoost, random forest and neural networks (up to 4 hidden layers), new causal features, strict chronological splits, long/short, long-only and short-only backtests.

Verdict

Optimisation adds less than one point of accuracy. Unfiltered, the Long & Short versions of the 5 candidates lose after costs. On daily candles most models call “up” every day, which just replicates staying long. None of the 5 validation-selected candidates passes the pre-set tests.

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Task, Data and Models

Predict the body of the next candle at the close of the current one, then trade exactly that candle

Prediction task (no look-ahead)



- **Target:** bullish if close(t+1) > open(t+1), bearish if lower; doji = miss
- **Data:** NQ futures (CME E-mini Nasdaq-100, $20 per point), TradeStation 1-min bars 2006–2025
- **Candles:** 1, 5, 15, 60 min and daily (one CME session, 17:00–16:00 CT)
- **Splits:** train 2006–Aug 2017 · validation → Jul 2021 · test Jul 2021–May 2025, purged at boundaries

Model families: original and optimised versions

1. **Logistic regression**
   weighted sum of features → probability of an up candle; simple, hard to overfit
2. **Random forest**
   averages hundreds of decision trees built on random data subsets
3. **XGBoost**
   trees added one by one, each correcting the previous errors; early-stopped on validation
4. **Neural network (MLP)**
   stacked layers of neurons learning non-linear patterns; 2 to 4 hidden layers tested

Features: 8 original price features (lagged returns, momentum, volatility) + 11 new ones (candle body and wicks, range, volume, realised volatility, RSI, moving-average gaps, time of day).

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Original vs Optimised Models

Optimisation changes out-of-sample accuracy by less than one point; deeper networks give no consistent gain

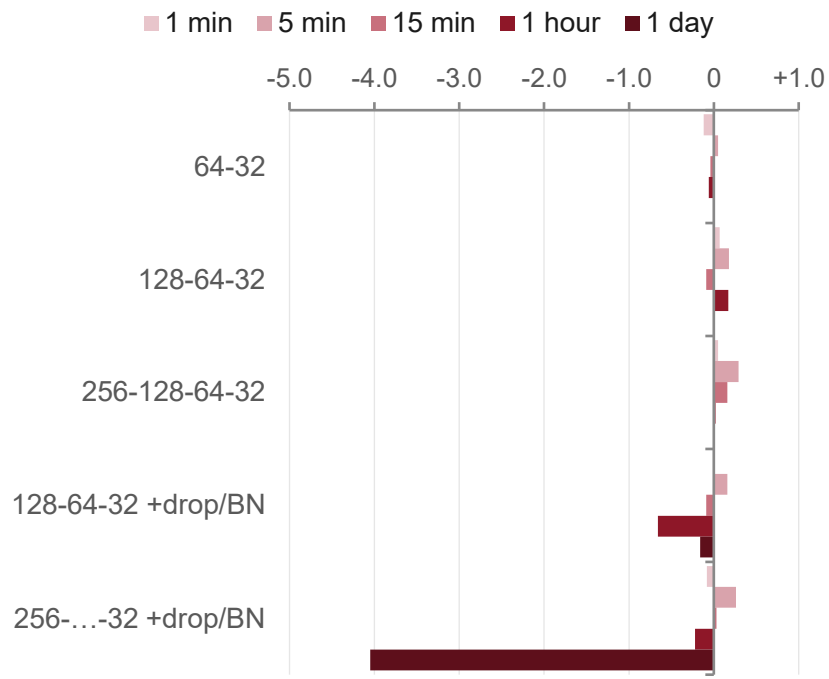
Test accuracy, all candles (original → optimised, change in pp)

Model	1 min	5 min	15 min	1 hour	1 day
Logistic regression	48.63% → 48.67% +0.04 pp	50.42% → 50.43% +0.01 pp	50.73% → 50.77% +0.05 pp	51.09% → 50.50% −0.59 pp	51.71% → 51.71% +0.00 pp
Random forest	48.68% → 48.69% +0.01 pp	50.49% → 50.47% −0.02 pp	50.88% → 50.79% −0.09 pp	51.11% → 51.69% +0.59 pp	54.21% → 54.21% +0.00 pp
XGBoost	48.63% → 48.63% −0.00 pp	50.50% → 50.46% −0.04 pp	50.67% → 50.96% +0.29 pp	50.77% → 51.76% +0.98 pp	51.56% → 51.56% +0.00 pp
Neural network (MLP)	48.66% → 48.71% +0.05 pp	**50.33% → 50.52% +0.18 pp**	50.94% → 50.85% −0.09 pp	51.57% → 51.74% +0.17 pp	54.21% → 54.21% +0.00 pp

Highlighted: best of the validation-selected models (one per family and candle), MLP 128-64-32 on 5 min: 51.45% on non-doji candles, +0.62 pp vs always-up. Across all 15 configurations, MLP 256-128-64-32 (5 min) shows +0.73 pp on test, descriptively; it was not selected on validation. Intervals are exploratory, not adjusted for the model search. Dojis count as misses.

Accuracy beats "always up" only at 1 and 5 minutes; from 15 minutes on, no model is reliably better

Neural networks: test accuracy vs original 64-32-16, pp



Early stopping halted every network after 5–17 epochs: validation loss stops improving almost immediately, a sign of a very weak signal. More neurons or layers gave no consistent gain; dropout/batch norm did not help.

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From Signals to Trades

Unfiltered, the Long & Short versions of the five validation-selected candidates lose after $5; intraday their gross edge is below $5 a trade

Trading rules (one NQ contract)

1 — Signal
model probability that candle t+1 is bullish

2 — Direction
Long & Short · Long only · Short only

3 — Filter (optional)
trade only if |P(up) − 0.5| ≥ threshold chosen on validation

4 — Execution
enter at open(t+1), exit at close(t+1); P&L = ±(close − open) × $20

5 — Costs
$5 per round trip ($2.50 open + $2.50 close); idealised: zero latency, zero slippage, 100% fills

Validation-chosen model per candle length, Long & Short, test period

Candle	Model	Trades	Gross / trade	Net	Filtered trades	Filtered gross / trade	Filtered net
1 min	MLP 256-128-64-32	1,370,692	−$0.17	**−$7.09M**	2,163	+$19.19	**+$30.7k**
5 min	MLP 128-64-32, lr 3e-4	273,340	+$0.42	**−$1.25M**	9,717	−$0.90	**−$57.4k**
15 min	MLP 128-64-32, lr 3e-4	90,189	−$0.53	**−$498.8k**	1,082	+$2.74	**−$2.4k**
1 hour	XGBoost d5, early stop	21,507	+$1.52	**−$74.9k**	17,419	+$4.56	**−$7.6k**
1 day	MLP 256-128-64-32 + dropout/BN	642	−$60.61	**−$42.1k**	575	−$40.09	**−$25.9k**

Gross / trade = average P&L per trade before the $5 commission. The filter (threshold chosen on validation) cuts trades by 10% to 99.8%; what remains is close to zero and unreliable.

Example, 5-minute reversal rule: +$391.3k gross over 273,340 trades becomes −$975.4k after commissions: $1.43 gross edge per trade vs $5.00 cost

Base case: idealised fills at the next open, zero latency, 100% fills, zero slippage, $5 per round trip

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Backtest by Model: Daily Candles

After costs 10 of 13 model × direction combinations are positive; 2 of 5 models call “up” on every day

Test period, no filter, one contract, $5 per round trip (1 day candles chosen on validation)

Model	Accuracy	Net: Long & Short	Net: Long only	Net: Short only	Trades	Win rate	Gross / trade	Net / trade	Max DD (session-end)
Logistic (original)	51.71%	+$110.0k	+$92.4k	+$17.6k	642	51.7%	+$176.31	+$171.31	−$70.3k
XGBoost (original)	51.56%	+$56.9k	+$66.1k	−$9.1k	642	51.6%	+$93.66	+$88.66	−$61.3k
Random forest d10	54.21%	+$73.0k	+$73.0k	no trades	642	54.2%	+$118.75	+$113.75	−$79.9k
MLP 64-32-16 (original)	54.21%	+$73.0k	+$73.0k	no trades	642	54.2%	+$118.75	+$113.75	−$79.9k
MLP 256-128-64-32 + dropout/BN	50.16%	−$42.1k	+$16.1k	−$58.2k	642	50.2%	−$60.61	−$65.61	−$100.9k
Always long	54.21%	—	+$73.0k	—	642	54.2%	+$118.75	+$113.75	−$79.9k

Trades, win rate, per-trade P&L and drawdown: Long & Short version (Long only for “always long”). Gross / trade is before the $5 commission. Highlighted: model of the pre-specified best strategy.

Staying long on every candle earned +$73.0k with no model; models that always call “up” replicate it exactly. 1 of 5 beat it with Long & Short; none of these has a daily-P&L 95% interval above zero

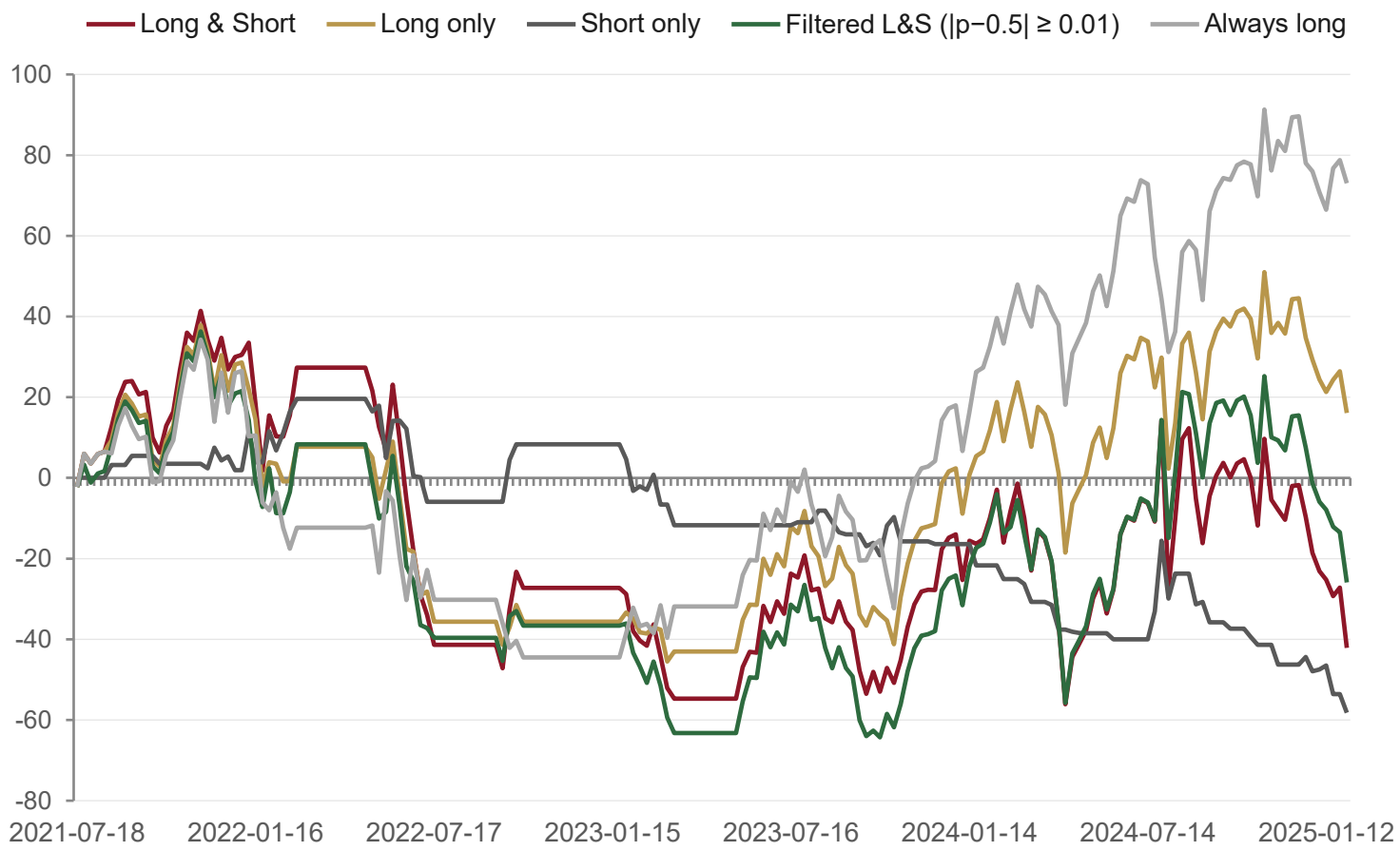
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Equity Curves: Long vs Short

Long-only beats short-only, but no direction beats simply staying long (+$73.0k)

Cumulative net P&L, test period, $k: MLP 256-128-64-32 + dropout/BN, 1 day candles



Long vs short contribution (no filter)

	Long only	Short only
Trades	520	122
Net P&L	+$16.1k	−$58.2k
Win rate	52.7%	39.3%
Gross / trade	+$35.89	−$471.93
Max DD, session-end	−$83.4k	−$84.1k

- The model calls “up” 81% of the time, so Long & Short is mostly a long position
- Short trades destroy value: −$58.2k net
- Long & Short: gross −$38.9k, commissions −$3.2k, net −$42.1k

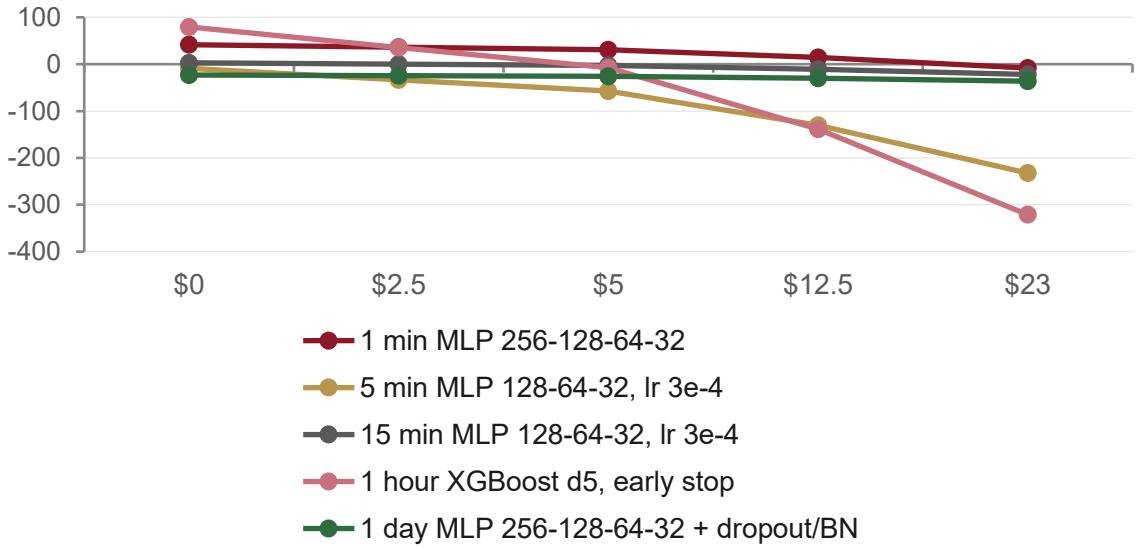
Base case: idealised fills at the next open, zero latency, 100% fills, zero slippage, $5 per round trip

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Robustness: Costs, Walk-Forward, In vs Out of Sample

Every candidate profitable on validation loses or shrinks on the test period

Test net P&L vs cost per round trip, filtered candidates (thousand USD)



$12.50 and $23 are assumed scenarios (one tick of spread plus ¼ or 1 tick of slippage per side), not measured execution costs.

Net P&L by sample and walk-forward years (filtered, $5)

Candle	Train	Validation	Test	Walk-forward years > 0	Walk-forward net
1 min	+$1.7k	+$48.8k	**+$30.7k**	2 of 4	+$3.7k
5 min	−$63.1k	+$81.3k	**−$57.4k**	1 of 4	−$16.7k
15 min	+$6.8k	+$51.0k	**−$2.4k**	1 of 4	−$65.5k
1 hour	+$63.2k	+$133.8k	**−$7.6k**	2 of 4	−$157.5k
1 day	+$102.6k	+$164.5k	**−$25.9k**	2 of 4 (2 n/e)	+$118.7k

Validation picks the best of 407 configurations, so its profit is optimistic by construction. Walk-forward covers these 5 candidates only: refitted each year on all prior data, threshold re-chosen on the preceding year; n/e = not evaluable (fewer than 200 validation trades, no threshold).

None of the 5 validation-selected candidates has a daily-P&L 95% interval above zero on the test, and none is profitable in at least 3 of 4 walk-forward years. Descriptively, 3 of 407 test configurations do (e.g. 1 min MLP 256-128-64-32 short only filtered, +$28.5k): found by searching, not independent confirmations.

Base case: idealised fills at the next open, zero latency, 100% fills, zero slippage, $5 per round trip

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Best Overall Strategy

No deployable strategy demonstrated under the pre-set criteria

Selected by the pre-specified rule
highest validation profit across all 407 configurations

Model	MLP 256-128-64-32 + dropout/BN
Candles · direction	1 day · Long & Short
Confidence filter	|P(up) − 0.5| ≥ 0.01
Test accuracy	50.16%
Net P&L (test)	**−$25.9k**
Trades · win rate	575 · 50.6%
Avg trade: gross · net	−$40.09 · −$45.09
Profit factor*	0.97
Max drawdown (session-end)	−$102.4k
Daily net 95% CI	−$351 to $267
Walk-forward years > 0	2 of 4 (2 n/e), +$118.7k

Highest test P&L among the five validation-selected candidates — descriptive only
picked by looking at the test: not a valid selection

Model	MLP 256-128-64-32
Candles · direction	1 min · Long & Short
Confidence filter	|P(up) − 0.5| ≥ 0.1
Test accuracy	48.71%
Net P&L (test)	**+$30.7k**
Trades · win rate	2,163 · 52.1%
Avg trade: gross · net	+$19.19 · +$14.19
Profit factor*	1.10
Max drawdown (session-end)	−$16.4k
Daily net 95% CI	−$11 to $85
Walk-forward years > 0	2 of 4, +$3.7k

* Profit factor = sum of winning trades ÷ |sum of losing trades|, both after commissions; 1.00 = winners exactly pay for losers. Avg trade gross = before the $5 commission. n/e = walk-forward year not evaluable.

Verdict: no deployable strategy demonstrated. None of the 5 validation-selected candidates passes the pre-set bar (test profit at $5 and $12.50, daily-P&L interval above zero, ≥ 3 of 4 profitable walk-forward years). 1 day walk-forward: +$118.7k on 2 evaluable years vs −$12.5k for always long on the same sessions; one path, not proof of an edge.

Base case: idealised fills at the next open, zero latency, 100% fills, zero slippage, $5 per round trip

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